MILITARY SPECIFICATION

COMPRESSOR, PISTON RING, OPEN CLAMP, ADJUSTABLE

1. SCOPE

1.1 This specification covers an open clamp adjustable type piston ring compressor.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-640	Steel, Carbon: Sheet and Strip
PPP-B-585	Boxes: Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-L-644	Lubricating-Oil, Preservative, Special
MIL-W-6858	Welding, Low Carbon Steels, Austenitic Steels, and Nickel Alloys, Spot and Seam
MIL-W-6873	Welding, Flash, Carbon and Alloy Steel
MIL-C-21968	Coating, Phosphate, Protective (for Iron and Steel)
MIL-S-18729	Steel Plate, Sheet, and Strip, Alloy, 4130, Aircraft Quality

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-120	Identification Marking of U. S. Military Property

AIR FORCE-NAVY AERONAUTICAL

AN350	Nut, Plain, Wing

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DRAWING

TRANSPORTATION CORPS

5120-B-1 Piston Ring Compressor, Open Clamp, Adjustable, Details and Assembly

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. - The ring compressor shall be of the adjustable, open-clamp, grip-operated type having the capability of compressing piston rings into the grooves of pistons 4 through 7 inches in diameter.

3.2 Materials. -

3.2.1 Bands and handles. - Bands and handles shall be of steel conforming to Specification MIL-S-18729.

3.2.2 Fastenings. - Fastenings shall be as shown on Drawing 5120-B-1.

3.2.2.1 Washer. - The flat washer shall be of carbon steel, cold-rolled half or full hard, conforming to Specification QQ-S-640.

3.2.2.2 Machine screw. - The machine screw shall be of steel, No. 10-32 with a class 2 thread.

3.2.2.3 Wing nut. -- The wing nut shall conform to part No. AN350-1032 as shown on Standard AN350.

3.3 Construction. -- The ring compressor shall be fabricated and assembled in conformance with Drawing 5120-B-1. The flash weld shall conform to Specification MIL-W-6873 and the spot welding to Specification MIL-W-6858. During assembly of the ring compressor, the knurled side of the flat washer shall be placed against the expansion band.

3.4 Finish. -- Ring compressor finish shall be as indicated on Drawing 5120-B-1. Finish of areas not indicated shall be equivalent to the rolled finish of the specified materials (see 3.2).

3.5 Marking. -- Marking of the ring compressor shall be on the inside of either handle, and in conformance with Standard MIL-STD-130 as follows:

Manufacturer's identification _____
FSN _____
U. S.

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3.6 Coating. -- Phosphate coating conforming to type B, class 1 of Specification MIL-C-12968 shall be applied to the entire compressor. Oil used for the supplementary petroleum base finish shall conform to Specification MIL-L-644. Coating of the wing nut may be omitted.

3.7 Workmanship. -- Workmanship shall be of the quality necessary to produce a compressor free from defects and blemishes affecting appearance or serviceability. Shearing and cutting shall be done neatly and accurately. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Upset metal of the flash weld shall be removed. Exposed parts shall have rounded corners and edges and be free of burrs or other safety hazards.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. -- Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Lot, inspection level, and acceptable quality level (AQL). -

4.1.1.1 Lot. -- A lot shall consist of 300 units or any fraction thereof. The samples shall be selected from the lot at random.

4.1.1.2 Inspection level and AQL. -- The inspection level shall be I, with AQL of 4.0 in conformance with Standard MIL-STD-105.

4.1.2 Rejection and retest. -- Failure of the ring compressors to conform to 4.2 shall be cause for the rejection of the entire lot. The rejected lot may be resubmitted only after the contractor has inspected the entire lot and corrected the cause for rejection. Before resubmission, details concerning previous rejection and action taken to correct the defects in the original, shall be furnished the Government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.2 Acceptance. - The samples obtained as indicated in 4.1.1 shall be inspected to determine that fabrication, assemblies, and finish conform to this specification. Adjustability of sample items shall be checked for minimum and maximum range (see 3.1).

4.3 Phosphate coating. -- Certification that the coating (see 3.6) has been applied in conformance with Specification MIL-C-12968 shall be furnished.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. --

5.1.1 Level A. -- Each ring compressor shall be individually packaged in accordance with method IC, without preservative, of Specification MIL-P-116.

5.1.2 Level B. -- Each ring compressor shall be individually packaged in accordance with method III of Specification MIL-P-116.

5.1.3 Level C. -- The ring compressor shall be preserved and packaged in accordance with standard commercial practice.

5.2 Packing. --

5.2.1 Level A. -- The ring compressors packaged as specified in 5.1 shall be packed in wood-cleated solid fiberboard, cleated plywood nailed wood, or wirebound boxes conforming to Specification PPP-B-591 (overseas type), PPP-B-601 (overseas type), PPP-B-621 (class 2), or PPP-B-585 (class 3). Gross weight shall not exceed 200 pounds.

5.2.2 Level B. -- The compressors preserved and packaged as specified in 5.1 shall be packed in wood-cleated solid fiberboard, cleated plywood, nailed wood, or wirebound boxes conforming to Specification PPP-B-591 (domestic type), PPP-B-601 (domestic type), PPP-B-621 (class I), or PPP-B-585 (class I). Gross weight shall not exceed 200 pounds.

5.2.3 Level C. --- The compressors shall be packed in a manner to insure acceptance by common carrier and safe delivery at destination at the lowest applicable rate.

5.3 Marking. -- In addition to any special marking required by the procurement documents, marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. -- The ring compressor is intended for use in com pressing piston rings into the grooves on pistons 4 through 7 inches in diameter, when installing pistons in cylinders.

6.2 Federal stock number. -- Federal stock number of this item is FSN 5120-289-8344.

6.3 Ordering data. -- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Records should be kept complete and available for at least 1 year (see 4.1).
(c) Whether preparation for delivery shall be at level A, B, or C (see 5.1 and 5.2)
(d) Whether special marking is required (see 5.3).

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6.4 Recommendations. - Recommendations to improve the ring compressor conforming to this specification are desired. Difficulties arising from materials, design, operation, or maintenance should be directed to the Chief of Transportation, Department of the Army, Washington 25, D. C. Manufacturers are particularly invited to submit suggestions for technical improvements and for decreasing costs. Recommendations should be in sufficient detail to unmistakably identify the item to be changed or the defect to be corrected.

Notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Interested activity:

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